

Comparison of all model outputs

Differences and similarities across the 4 fusion outputs (rule-based + partner score per model).

1. Summary per model

Model	Flagged count	% of total	Threshold (approx.)
ABOD k60 pct	3,071	5.00%	0.6498
clustered LOF	3,071	5.00%	0.4380
full PCA LOF k240	3,071	5.00%	0.3653
Isolation Forest c05 n100	3,071	5.00%	0.4444

Insight: Each row is one model (rule-based + that model's anomaly score). Flagged count is always the top 5% of customers for that model. Thresholds can differ because the partner scores (e.g. Isolation Forest vs LOF) have different scales and distributions; a higher threshold means the model needs a higher combined score to reach the top 5%. Comparing counts and thresholds shows which models are more or less conservative in practice.

2. Overlap: how many models flagged each customer

# models that flagged	Customer count	% of total
1	3,258	5.31%
2	787	1.28%
3	1,232	2.01%
4	939	1.53%

Insight: Row 'k' is the number of customers flagged by exactly k models. Customers flagged by only one model are unique to that model's view of risk; customers flagged by all 4 models are the strongest consensus alerts. The distribution tells you how much the models agree: if most customers are in the '1' or '4' column, models tend to either disagree or fully agree; if many sit in the middle, there is partial overlap across model pairs.

3. Consistently flagged (by all 4 models)

Count: 939 customers were flagged by every model. These are the strongest candidates for review.

Insight: These 939 customers rank in the top 5% under every fusion (rule + each partner score). They should be prioritised first: the rule-based logic and every anomaly model agree they are high risk. Use the per-model explanation files to see why each was flagged; the reasons will often emphasise both rule-driven and anomaly-driven factors.

4. Pairwise overlap (customers flagged by both models)

ABOD k60 pct clustered LO full PCA LOF Isolation Fo

ABOD k60 pct	3071	1130	2391	2274
clustered LOF	1130	3071	1069	1109
full PCA LOF k	2391	1069	3071	2144
Isolation Fore	2274	1109	2144	3071

Insight: Each cell (A, B) is the number of customers flagged by both model A and model B. The diagonal is the total flagged per model. High off-diagonal values mean those two models often flag the same customers; lower values mean they focus on different subsets. Comparing pairs helps you see which models are most aligned (e.g. LOF variants vs Isolation Forest) and which add distinct signal.

5. Similarities and differences

Similarities

All outputs use the same rule-based score (weight 0.7) and the same top-5% rule; only the partner (anomaly) score (weight 0.3) changes per model. The 939 consistently flagged customers are the core high-risk set on which all models agree and should be reviewed first. Flag counts are similar in size because each model flags 5% of the population; what changes is which customers make up that 5%.

Differences

Each model uses a different anomaly or ML score (e.g. Isolation Forest vs LOF variants), so the combined risk score and the effective threshold differ. That is why some models have a higher or lower threshold: the scale of the partner score affects where the top 5% cutoff falls. Customers flagged by only one or two models are more model-specific; those flagged by four or all models are more robust candidates. Use the pairwise overlap table to see which model pairs agree most, and use the per-model reports to interpret each model's distribution in detail.